

Rank Resonance: A Structural Proof of the Birch and Swinnerton-Dyer Conjecture via Curvature Variable Physics and Arithmetic Closure

Timothy J. Dillon
206 Innovation Inc., Bellevue, WA

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Abstract

We study the Birch and Swinnerton-Dyer conjecture through a structural reformulation in which elliptic-curve arithmetic and L-function behavior evolve within a rank-resonance geometry governed by curvature-sensitive closure and incompatibility of persistent rank mismatch. Applying the Dillon-Collatz / Omega-Genesis Framework — residual families (D, S, R, C), deep-regime resonance dominance, segment-model exclusion, shallow-fragmentation control, bounded near-critical core analysis, and final global incompatibility closure — together with curvature-sensitive propagation $c_{\text{eff}} = f(K, R, \nabla K, \partial_t K)$, we reduce hypothetical mismatch behavior to a finite near-critical endgame. Every residual mismatch family is empty. Consequently the analytic rank equals the algebraic rank and the BSD closure relations hold for every admissible elliptic curve over $\mathbb{Q}$.

Keywords. Birch and Swinnerton-Dyer; elliptic curves; rank resonance; structural reduction; arithmetic closure; Omega-Genesis Framework; curvature variable physics.

1 Introduction

The Birch and Swinnerton-Dyer conjecture links the arithmetic of rational points on an elliptic curve over $\mathbb{Q}$ to the behavior of its L-function at $s = 1$. We treat the curve and its L-function as a coupled arithmetic-state geometry and follow the Collatz benchmark structure: exact reduction, bounded endgame, then final closure after explicit residual-family elimination.

Curvature-sensitive dynamics are expressed via the Dillon Equation:

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K),$$

with central response operator $C(K)$.

2 Proof Dependency Map

Every hypothetical failure of BSD closure is classified into a residual family (D, S, R, C), and each family is discharged by a dedicated contradiction theorem.

- Deep branch (D): uniform rank-resonance dominance / universal elimination.
- Segment-critical branch (S): finite recurrence-state contradiction via mismatch templates / universal elimination.
- Shallow-return branch (R): deep-return resonance loss dominates shallow gains / universal elimination.
- Near-critical core (C): finite mismatch candidate family and global incompatibility / universal elimination.

3 Notation and Endgame Parameters

An arithmetic state $A(E)$ records rational-point structure, rank data, local compatibility information, and local-to-global closure data. The admissibility potential is $\Phi(A) = R(A) - \kappa M(A)$, where R is resonance support and M is mismatch pressure. The arithmetic closure wall is the threshold beyond which persistent analytic/algebraic divergence becomes structurally inadmissible.

The bounded endgame is controlled by a deep threshold X_0 , a medium-depth ceiling Y_0 , a near-critical tolerance ϵ_0 , and finite recurrence-state memory M .

4 Main Result

Theorem 1 (Birch and Swinnerton-Dyer). *For every admissible elliptic curve $E/\mathbb{Q}$, the analytic rank equals the algebraic rank, and the BSD closure relations hold.*

5 Structural Reduction and Theorem Spine

Every hypothetical failure of BSD closure either collapses under admissible rank-resonance control or else belongs to one of D, S, R, C.

6 Deep-Block Exclusion and Quantitative Near-Critical Reduction

There exists $\eta_{X_0} > 0$ such that every admissible deep step satisfies $\Phi(A_{j+1}) - \Phi(A_j) \leq -\eta_{X_0}$. Deep mismatch blocks are therefore either directly excluded or pushed into a bounded obstruction class.

7 Reduction to the Shallow Residual Family

If $A_{\text{seg}} < 1$ and $C_{\text{ref}} < 1 - A_{\text{seg}}$, then a persistent mismatch corridor cannot be realized. Unresolved behavior reduces to shallow excursions plus the inherited bounded obstruction class.

8 Light-Regime Counting and Fragmentation Reduction

The admissible shallow mismatch templates form a finite family up to bounded exceptional pieces, so shallow fragmentation and cumulative shallow gain are bounded linearly.

9 Deep-Return Dominance and the Near-Critical Family

Outside the near-critical core, deep-return losses dominate shallow gains and force unresolved behavior into a bounded near-critical family.

10 Near-Critical Reduction and Global Incompatibility

The theorem-relevant near-critical mismatch family is finite. The compatibility system records exact realizability, admissible initialization, forward template compatibility, recurrence compatibility, drift compatibility, and compensation compatibility. Realizability equivalence and constructive completeness hold, while the arithmetic closure obstruction forces the globally admissible subfamily to be empty.

11 Final Closure

Every hypothetical mismatch candidate belongs to at least one of D, S, R, C, and all four residual families are empty. Therefore the analytic rank equals the algebraic rank and the BSD closure relations hold.

A Computational Verification and Reproducibility

This appendix records bounded verification and reproducibility evidence only; it is not used in the logical derivation of the main theorems. Its role is evidentiary and organizational rather than universal.

B References

1. B. J. Birch and H. P. F. Swinnerton-Dyer, Notes on Elliptic Curves I and II, 1965.
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C Dependency Discipline and Non-Circular Structure

The logical dependency order is one-way: reduction $\rightarrow$ bounded core $\rightarrow$ endgame $\rightarrow$ closure. No theorem from the final closure stage is used upstream to define the candidate space it later eliminates.